

Day-6

29. Find names of instructors with salary greater than that of some (at least one) instructor in the Biology department.

Query:

```
SELECT name from instructor WHERE salary > some  
(SELECT salary FROM instructor WHERE dept_name = 'Biology');
```

name

Bawa

Yazdi

Wieland

D'Agostino

Liley

Atanassov

Moreira

Gustafsson

Bourrier

Bondi

Soisalon-Soininen

Arias

Murata

Tung

Luo

Romero

Lent

Sarkar

Shuming

Bancilhon

Dusserre

Desyl

Jaekel

McKinnon

Mingoz

30. Find names of all instructors that have salary value greater than of each instructor in the Biology department.

Query :

```
SELECT name  
FROM instructor  
WHERE salary > all(  
    SELECT salary  
    FROM instructor  
    WHERE dept_name = 'Biology');
```

name

Yazdi

Wieland
Liley
Atanassov
Gustafsson
Bourrier
Bondi
Arias
Luo
Romero
Lent
Sarkar
Shuming
Bancilhon
Jaekel
McKinnon
Mingoz
Pimenta
Sullivan
Voronina
Kenje
Mahmoud
Levine
Bietzk
Sakurai

31. Find the departments that have the highest average salary.

Query:

```
SELECT dept_name
FROM instructor
GROUP BY dept_name
HAVING AVG(salary) >= ALL(SELECT AVG(salary)
                           FROM instructor
                           GROUP BY dept_name);
```

Physics

32. Find all courses taught in both the Fall 2017 semester and in the Spring 2018 semester.

Query:

```
Select course_id
from section as S
where semester = 'Fall' and year = 2017 and
      exists (select *
              from section as T
              where semester = 'Spring' and year= 2018
              and S.course_id = T.course_id);
```

33. Find all students who have taken all courses offered in the Biology department.

Query:

```
select distinct S.ID, S.name
from student as S
where not exists ( (select course_id
                    from course
                    where dept_name = 'Biology')
except
(select T.course_id
 from takes as T
 where S.ID = T.ID));
```

34. Find all courses taught in both the Fall 2007 semester and in the Spring 2008 semester.

Query :

```
SELECT course_id
FROM section
WHERE semester = 'Fall' and year = 2007 and
EXISTS(SELECT course_id
        FROM section
        WHERE semester ='Spring' and year = 2008 );
course_id
893
489
612
258
468
949
```